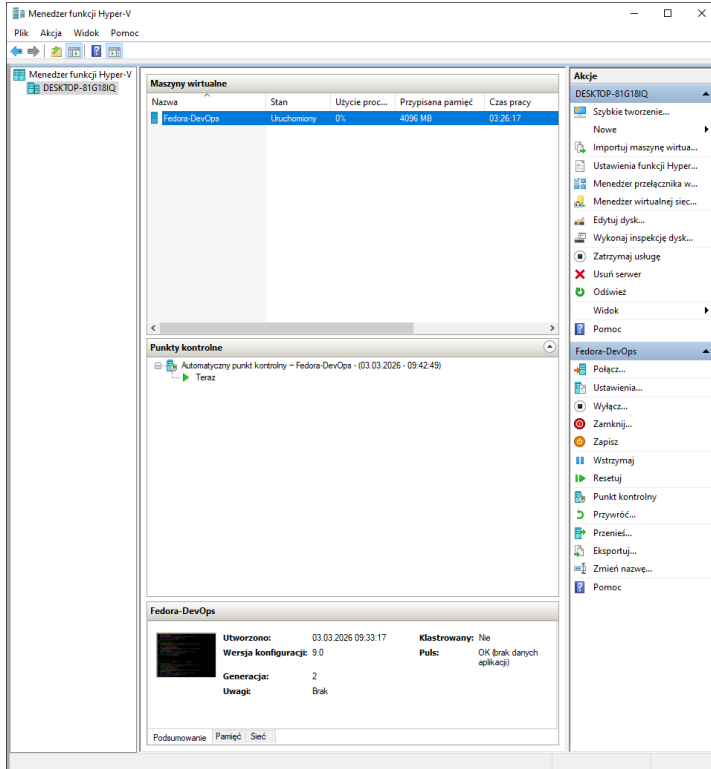


# Sprawozdanie LAB 1 - 4

Szymon Krzysztofik

## LAB 1

Konfiguracja maszyny wirtualnej. Używam do tego Hyper-V i systemu Fedora Server.



Stworzyłem klucze szyfrowane ed25519 z hasłem oraz ECDSA bez hasła, ich publiczne odpowiedniki widać na screenie. Skonfigurowałem także środowisko git.

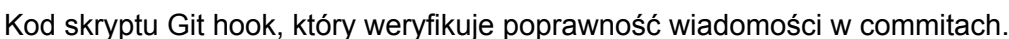
```
PS C:\Users\Szymon> ssh skrzysztofik@172.17.191.2
skrzysztofik@172.17.191.2's password:

1 urządzenie ma dostępną aktualizację oprogramowania sprzętowego.
Polecenie „fwupdmgrr get-upgrades” wyświetli więcej informacji.

Web console: https://localhost:9090/ or https://172.17.191.2:9090/

Last login: Tue Mar  3 11:39:10 2026 from 172.17.176.1
skrzysztofik@localhost:~$ cat ~/.ssh/id_ed25519_haslo.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIP3e8J8x4IFYuLR7Fi6KexpG5PZFQeSXJxPzp45JTQp szymon.krzysz
zt@gmail.com
skrzysztofik@localhost:~$ cat ~/.ssh/id_ecdsa_bez_hasla.pub
ecdsa-sha2-nistp521 AAAAE2VjZHNhLXNoYTItbmlzdHA1MjEAAAABbmLzdHA1MjEAAAACFBACtblaCor5T0Tc1fbRKp
hf0ep+iD8NCWfMWFds4Wuxcvj9hwCF7RH9vBV0W4rbtWbnKDW2sf06OnNTVFweju0ptVwGH4vKtp1q/kZB91Hy96UeIdS
H0n/2Ng8i8N2pGKLugsGLhd0aCLE7xnFLIUxTqjz9nPFTLFCF/5QRXLK0kiuYZqQ== szymon.krzyszzt@gmail.com
skrzysztofik@localhost:~$ git --version
git version 2.53.0
skrzysztofik@localhost:~$ ls
MD02026s_ITE
```

Konfiguracja klucza SSH jako metody dostępu do serwisu GitHub.



```

$ commit-msg X image.png sprawozdanie1.md M
MDO2026s ITE > ITE > GCL3 > SK419875 > $ commit-msg
1  #!/bin/bash
2  commit_msg=$(cat $1)
3  required_start="SK419875"
4  if [[ ! $commit_msg =~ ^$required_start ]]; then
5      echo "ERROR: Commit message must start with $required_start"
6      exit 1
7  fi

skrzysztofik@localhost:~/MDO2026s ITE/ITE/GCL3/SK419875/lab1$ git add .
skrzysztofik@localhost:~/MDO2026s ITE/ITE/GCL3/SK419875/lab1$ git commit -m "sprawozdanie 1"
cat: 1: Nie ma takiego pliku ani katalogu
ERROR: Commit message must start with SK419875

```

## LAB 2

Ściągnięte obrazy hello-world, busybox, ubuntu, mariadb, runtime, aspnet i sdk dla Microsoft .NET i ich kody wyjścia. Mariadb potrzebuje hasła, którego nie wpisałem - dlatego zakończyło się jako 1 (błąd)

```

❯ skrzysztofik@localhost:~$ docker run mariadb
2026-03-10 23:29:43+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:12.2.2+maria-ubu2404 started.
2026-03-10 23:29:43+00:00 [Warn] [Entrypoint]: /sys/fs/cgroup///memory.pressure not writable, functionality unavailable to MariaDB
2026-03-10 23:29:43+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-03-10 23:29:43+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:12.2.2+maria-ubu2404 started.
2026-03-10 23:29:43+00:00 [ERROR] [Entrypoint]: Database is uninitialized and password option is not specified
You need to specify one of MARIADB_ROOT_PASSWORD, MARIADB_ROOT_PASSWORD_HASH, MARIADB_ALLOW_EMPTY_ROOT_PASSWORD and MARIADB_RANDOM_ROOT_PASSWORD
❯ skrzysztofik@localhost:~$ docker run aspnet
Unable to find image 'aspnet:latest' locally
docker: Error response from daemon: pull access denied for aspnet, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
❯ skrzysztofik@localhost:~$ docker run mcr.microsoft.com/dotnet/aspnet
❯ skrzysztofik@localhost:~$ docker run mcr.microsoft.com/dotnet/runtime
❯ skrzysztofik@localhost:~$ docker run mcr.microsoft.com/dotnet/sdk
❯ skrzysztofik@localhost:~$ docker ps -a

```

| CONTAINER ID | IMAGE                            | COMMAND                  | CREATED            | STATUS                        | PORTS | NAMES                 |
|--------------|----------------------------------|--------------------------|--------------------|-------------------------------|-------|-----------------------|
| 23cc732b735f | mcr.microsoft.com/dotnet/sdk     | "/bin/bash"              | 32 seconds ago     | Exited (0) 31 seconds ago     |       | zen_moore             |
| 0a50cea9e501 | mcr.microsoft.com/dotnet/runtime | "/bin/bash"              | 50 seconds ago     | Exited (0) 50 seconds ago     |       | distracted_hofstadter |
| be3c66889946 | mcr.microsoft.com/dotnet/aspnet  | "/bin/bash"              | About a minute ago | Exited (0) About a minute ago |       | eloquent_mahavira     |
| 1dfc8530b7c3 | mariadb                          | "docker-entrypoint.s..." | About a minute ago | Exited (1) About a minute ago |       | dreamy_hopper         |
| 7ec3934c2991 | ubuntu                           | "bash"                   | 12 minutes ago     | Exited (0) 18 minutes ago     |       | loving_curran         |
| 8df456940ddb | ubuntu                           | "bash"                   | 22 minutes ago     | Exited (0) 15 minutes ago     |       | clever_shtenn         |
| c6f5b2deaf48 | busybox                          | "sh"                     | 23 minutes ago     | Exited (0) 22 minutes ago     |       | dazzling_swirles      |
| 9de1c199ff5b | busybox                          | "sh"                     | 24 minutes ago     | Exited (0) 24 minutes ago     |       | practical_herschel    |
| 6dfbb7b1201d | hello-world                      | "/hello"                 | 25 minutes ago     | Exited (0) 25 minutes ago     |       | wizardly_lederberg    |
| ea50b6421f1a | busybox                          | "sh"                     | 15 hours ago       | Exited (0) 15 hours ago       |       | gracious_mayer        |
| 7d9640181b35 | hello-world                      | "/hello"                 | 15 hours ago       | Exited (0) 15 hours ago       |       | dreamy_ride           |

```

❯ skrzysztofik@localhost:~$

```

Uruchomienie konteneru z obrazu busybox interaktywnie. Wersja to 1.37.0.

```

❯ skrzysztofik@localhost:~$ docker run -it busybox sh
/ # busybox
BusyBox v1.37.0 (2024-09-26 21:31:42 UTC) multi-call binary.
BusyBox is copyrighted by many authors between 1998-2015.
Licensed under GPLv2. See source distribution for detailed
copyright notices.

Usage: busybox [function [arguments]...]
or: busybox --list[-full]
or: busybox --show SCRIPT
or: busybox --install [-s] [DIR]
or: function [arguments]...

BusyBox is a multi-call binary that combines many common Unix
utilities into a single executable. Most people will create a
link to busybox for each function they wish to use and BusyBox
will act like whatever it was invoked as.

Currently defined functions:
l, ll, acpid, add-shell, addgroup, adduser, adjtimex, ar, arch, arp, arping, ascii, ash, awk, base32, base64, basename,
bc, beep, blkdiscard, blkid, blockdev, bootchartd, brctl, bunzip2, bzip2, cal, cat, chat, chatter, chgrp, chmod,
chown, chpasswd, chpst, chroot, chrt, chvt, cksum, clear, cmp, comm, conspy, cp, cpio, crc32, crond, crontab, cryptpw,
cttyhack, cut, date, dc, dd, deallocvt, delgroup, deluser, depmod, devmem, df, dhcprelay, diff, dirname, dmesg, dnsd,
dnsdomainname, dos2unix, dpkg, dpkg-deb, du, dumpkmap, dumpleases, echo, ed, egrep, eject, env, envdir, envuidgid,
etherwake, expand, expr, factor, fakeidentd, fallocafe, false, fatattr, fbset, fbsplash, fdflush, fdformat, fdisk,
fgconsole, fgrep, find, findfs, flock, fold, free, freeramdisk, fsck, fsck.minix, fsfreeze, fstrim, fsync, ftpd,
ftppget, ftpput, fuser, getfatattr, getopt, getty, grep, groups, gunzip, gzip, halt, hd, hdparm, head, hexdump, hexedit,
hostid, hostname, httpd, hush, hwclock, i2cdetect, i2cdump, i2cget, i2cset, i2ctransfer, id, ifconfig, ifdown,
ifenslave, ifplugd, ifup, inetd, init, inotifyd, insmod, install, ionice, iostat, ip, ipaddr, ipcalc, ipcrm, ipcs,
iplink, ipm6, iproute, iprule, iptunnel, kbd_mode, kill, killall, killall5, klogd, last, less, link, linux32,
linux64, linuxrc, ln, loadfont, loadkmap, logger, login, logname, logread, losetup, lpd, lpq, lpr, ls, lsattr, lsmod,
lsuf, lspci, lsscsi, lsusb, lzcat, lzma, lzop, makedevs, makemime, man, md5sum, mdev, mesg, microcom, mim, mkdir,
mkdosfs, mke2fs, mkfifo, mkfs.ext2, mkfs.minix, mkfs.vfat, mkmod, mkpasswd, mkswap, mktemp, modinfo, modprobe, more,
mount, mountpoint, mpstat, mt, mv, nameif, nanddump, nandwrite, nbd-client, nc, netstat, nice, nl, nmeter, nohup,
nologin, nproc, nsenter, nslookup, ntpd, od, openvt, partprobe, passwd, paste, patch, pgrep, pidof, ping, ping6,
pipe_progress, pivot_root, pkill, pmap, popmaildir, poweroff, powertop, printenv, printf, ps, pscan, pstree, pwd, pwdx,
raidautorun, rdate, rdev, readahead, readlink, readprofile, realpath, reboot, reformime, remove-shell, renice, reset,
resize, resume, rev, rm, rmdir, rmmmod, route, rpm, rpm2cpio, rtcwake, run-init, run-parts, runlevel, runsv, runsvdir,
rx, script, scriptreplay, sed, seedrng, sendmail, seq, setarch, setconsole, setfatattr, setfont, setkeycodes, setlogcons,
setpriv, setserial, setsid, setuidgid, sh, sha1sum, sha256sum, sha3sum, sha512sum, showkey, shred, shuf, slattach,
sleep, smemcap, softlimit, sort, split, ssl_client, start-stop-daemon, stat, strings, stty, su, sulogin, sum, sv, svc,
svlogd, svok, swapoff, swapon, switch_root, sync, sysctl, syslogd, tac, tail, tar, taskset, tc, tcpsvd, tee, telnet,
telnetd, test, tftp, tftpd, time, timeout, top, touch, tr, traceroute, traceroute6, tree, true, truncate, ts, tsort,
tty, ttysize, tuncit, ubiattach, ubidetach, ubiimvol, ubirename, ubirmvol, ubirsvol, ubiupdatevol, udhcpc, udhcpc6,
udhcpd, udpsvd, uevent, umount, uname, unexpand, uniq, unix2dos, unlink, unlzma, unshare, unxz, unzip, uptime, users,
usleep, uuencode, uunencode, vconfig, vi, vlock, volname, w, wall, watch, watchdog, wc, wget, which, who, whoami, whois,
xargs, xxd, xz, xzcat, yes, zcat, zcip
/ #

```

Uruchomienie "systemu w kontenerze", PID1 w kontenerze - bash. Zaktualizowanie pakietów.

```

Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2025 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2285 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.6 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1260 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1899 kB]
Fetched 36.6 MB in 3s (11.0 MB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
4 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@0df456940ddb:/# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  gcc-14-base libgcc-s1 libgnutls30t64 libstdc++6
4 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 1923 kB of archives.
After this operation, 9216 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-14-base amd64 14.2.0-4ubuntu2~24.04.1 [51.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++6 amd64 14.2.0-4ubuntu2~24.04.1 [792 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc-s1 amd64 14.2.0-4ubuntu2~24.04.1 [78.4 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgnutls30t64 amd64 3.8.3-1.1ubuntu3.5 [1001 kB]
Fetched 1923 kB in 1s (3696 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../gcc-14-base_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking gcc-14-base:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Setting up gcc-14-base:amd64 (14.2.0-4ubuntu2~24.04.1) ...
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libstdc++6_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libstdc++6:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Setting up libstdc++6:amd64 (14.2.0-4ubuntu2~24.04.1) ...
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libgcc-s1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libgcc-s1:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Setting up libgcc-s1:amd64 (14.2.0-4ubuntu2~24.04.1) ...
(Reading database ... 4381 files and directories currently installed.)
Preparing to unpack .../libgnutls30t64_3.8.3-1.1ubuntu3.5_amd64.deb ...
Unpacking libgnutls30t64:amd64 (3.8.3-1.1ubuntu3.5) over (3.8.3-1.1ubuntu3.4) ...
Setting up libgnutls30t64:amd64 (3.8.3-1.1ubuntu3.5) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
root@0df456940ddb:/#

```

Stan: Uruchomiony

```

root@0df456940ddb:/# apt install procps -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
procps is already the newest version (2:4.0.4-4ubuntu3.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@0df456940ddb:/# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1  4596  4100 pts/0    Ss   23:09   0:00 bash
root      212  0.0  0.1  7896  4172 pts/0    R+   23:11   0:00 ps aux
root@0df456940ddb:/#

```

Stan: Uruchomiony

```

DEBUG CONSOLE  PORTS  PROBLEMS  OUTPUT  TERMINAL
● skrzysztofik@localhost:~$ ps aux | grep docker
root         890  0.1  2.4 1729412 99664 ?        Ssl  Mar10   0:03 /usr/bin/dockerd -H fd:// --containerd=/run/containerd
/containers.sock --selinux-enabled --userland-proxy-path /usr/bin/docker-proxy --init-path /usr/bin/tini-static
skrzysz+  2591  0.0  0.6 1427308 25744 tty1      Sl+  00:18   0:00 docker run -it ubuntu bash
skrzysz+  2702  0.0  0.0   6560  2368 pts/3    St+  00:19   0:00 grep --color=auto docker
○ skrzysztofik@localhost:~$

```

Plik Dockerfile i sklonowanie w nim naszego repozytorium.

```
ITE > GCL3 > SK419875 > lab2 > Dockerfile > FROM

1 FROM ubuntu:latest
2
3 RUN apt-get update && apt-get install -y git
4
5 WORKDIR /workspace
6
7 RUN git clone https://github.com/InzynieriaOprogramowaniaAGH/MD02026s ITE.git .
8
9 CMD ["/bin/bash"]
10

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875$ docker build -t sk419875-lab2 .
[+] Building 33.1s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 283B                                0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest   0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/4] FROM docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4 0.0s
=> [2/4] RUN apt-get update && apt-get install -y git              13.8s
=> [3/4] WORKDIR /workspace                                         0.1s
=> [4/4] RUN git clone https://github.com/InzynieriaOprogramowaniaAGH/MD02026s ITE.git . 2.0s
=> exporting to image                                              16.8s
=> => exporting layers                                              15.1s
=> => exporting manifest sha256:db9d96d3012bc2635f7bd42e1015673b1283b3905cafb0b4d69db6c07bc12d2 0.0s
=> => exporting config sha256:ad0b453307a6a6ae53258f10f07de96d9997174a76e7bf0e7bc46bde68e2c88 0.0s
=> => exporting attestation manifest sha256:7b384f84297f00d99f680c9ba95be06f67880afe77d952ded7f0c7e65f973ce5 0.0s
=> => exporting manifest list sha256:91f1091af50a7ecff097bcf3ae0199d9beb154d4ea6981bb87f872e62567f245 0.0s
=> => naming to docker.io/library/sk419875-lab2:latest            0.0s
=> => unpacking to docker.io/library/sk419875-lab2:latest         1.6s
skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875$ docker run -it sk419875-lab2
Unable to find image 'sk419875-lab2:latest' locally
docker: Error response from daemon: pull access denied for sk419875-lab2, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875$ ls
commit-msg Dockerfile lab1
skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875$ docker run -it sk419875-lab2
root@a7716499e3de:/workspace# ls
README.md READMEs
root@a7716499e3de:/workspace#

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS   NAMES
a7716499e3de  sk419875-lab2  "/bin/bash"             2 minutes ago  Exited (0)   6 seconds ago          dreamy_poitras
23cc732b735f  mcr.microsoft.com/dotnet/sdk  "/bin/bash"             17 minutes ago  Exited (0)   17 minutes ago          zen_moore
0a50cea9e501  mcr.microsoft.com/dotnet/runtime  "/bin/bash"             17 minutes ago  Exited (0)   17 minutes ago          distracted_hofstadter
be3c66889946  mcr.microsoft.com/dotnet/aspnet  "/bin/bash"             18 minutes ago  Exited (0)   18 minutes ago          eloquent_mahavira
1dfc8530b7c3  mariadb       "docker-entrypoint.s..." 18 minutes ago  Exited (1)   18 minutes ago          dreamy_hopper
7ec3934c2991  ubuntu        "bash"                  29 minutes ago  Exited (0)   27 minutes ago          loving_curran
8df456940ddb  ubuntu        "bash"                  39 minutes ago  Exited (0)   32 minutes ago          clever_shtern
c6f5b2deaf48  busybox       "sh"                    41 minutes ago  Exited (0)   39 minutes ago          dazzling_swirls
9de1c199ff5b  busybox       "sh"                    41 minutes ago  Exited (0)   41 minutes ago          practical_herschel
6dfbb7b1201d  hello-world   "/hello"                42 minutes ago  Exited (0)   42 minutes ago          wizardly_lederberg
ea50b642f1fa  busybox       "sh"                    15 hours ago   Exited (0)   15 hours ago          gracious_mayer
7d9640181b35  hello-world   "/hello"                15 hours ago   Exited (0)   15 hours ago          dreamy_ride
```

Następnie oczyściłem środowisko za pomocą:  
**docker container prune**  
**docker image prune -a**  
ale niestety nie mam z tego zrzutów ekranu.

## LAB 3

Wybrałem oprogramowanie Marked.js - jest to lekka, szybka i bardzo popularna biblioteka JavaScript służąca do konwersji tekstu napisanego w języku Markdown na kod HTML. Poniżej znajduje się build i testy w kontenerze przez terminal.

```
✓ should use walkTokens (0.332901ms)
✓ should use options from extension (0.630001ms)
✓ should call all walkTokens in reverse order (0.435501ms)
✓ should use last extension function and not override others (0.539301ms)
✓ should use previous extension when returning false (0.597402ms)
✓ should get options with this.options (0.397101ms)
✓ use extension (40.828703ms)
  ✖ walkTokens
    ✓ should walk over every token (4.771012ms)
    ✓ should assign marked to `this` (0.314801ms)
    ✓ should wait for async `walkTokens` function (161.550708ms)
    ✓ should return promise if async and no walkTokens function (0.972602ms)
  ✓ walkTokens (167.941025ms)
✓ marked unit (241.504111ms)
i tests 165
i suites 36
i pass 165
i fail 0
i cancelled 0
i skipped 0
i todo 0
i duration_ms 1657.320694

> marked@17.0.5 test:umd
> node test/umd-test.js

> marked@17.0.5 test:cjs
> node test/cjs-test.cjs

> marked@17.0.5 test:types
> tsc --project tsconfig-type-test.json && attw -P --entrypoints . --profile esm-only

marked v17.0.5

Build tools:
- @arethetypeswrong/cli@^0.18.2
- typescript@5.9.3
- esbuild@^0.27.4
```

Dwa pliki Dockerfile: pierwszy automatyzujący proces pobierania zależności i budowania, a drugi bazujący na nim i wykonujący wyłącznie testy.

ITE > GCL3 > SK419875 > lab3 > Dockerfile > ...

```
1 FROM node:20
2
3 WORKDIR /app
4
5 RUN git clone https://github.com/markedjs/marked.git .
6
7 RUN npm install
8
9 RUN npm run build
```

ITE > GCL3 > SK419875 > lab3 > Dockerfile.test > FROM

```
1 FROM marked-build:latest
2
3 CMD ["npm", "test"]
```

Dowód, że przygotowany kontener wdraża się i pracuje poprawnie.

```
● skrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab3$ docker build -t marked-build .
[+] Building 68.1s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 214B                             0.0s
=> [internal] load metadata for docker.io/library/node:20       0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/5] FROM docker.io/library/node:20@sha256:e391c5561646193be929e3fe55d27234f40a01d867d72b 0.2s
=> => resolve docker.io/library/node:20@sha256:e391c5561646193be929e3fe55d27234f40a01d867d72b 0.0s
=> [2/5] WORKDIR /app                                           0.0s
=> [3/5] RUN git clone https://github.com/markedjs/marked.git . 5.1s
=> [4/5] RUN npm install                                        15.7s
=> [5/5] RUN npm run build                                       17.3s
=> exporting to image                                           29.4s
=> => exporting layers                                          24.8s
=> => exporting manifest sha256:9ba89febd37cbabca6bfe6a438e57710563c55d1af9a468dfefdcc097afdf 0.0s
=> => exporting config sha256:0ed7272a9db305c21657f2d1f758d1f8a2e98b15ffa97953045d5582b39915d 0.0s
=> => exporting attestation manifest sha256:ce12299b41bfef9aa2ba9169ae92158b3ac3900f5393b5165 0.0s
=> => exporting manifest list sha256:c4010b4020e20ddc86c806657691634942097173221dc20dfa9663dc 0.0s
=> => naming to docker.io/library/marked-build:latest          0.0s
=> => unpacking to docker.io/library/marked-build:latest       4.5s
○ skrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab3$
```

```

● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$ docker build -t marked-test -f Dockerfile.test .
[+] Building 0.5s (5/5) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile.test        0.0s
=> => transferring dockerfile: 146B                             0.0s
=> [internal] load metadata for docker.io/library/marked-build:latest 0.1s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/1] FROM docker.io/library/marked-build:latest@sha256:c4010b4020e20ddc86c806657691634942097173221dc20dfa966 0.2s
=> => resolve docker.io/library/marked-build:latest@sha256:c4010b4020e20ddc86c806657691634942097173221dc20dfa966 0.0s
=> exporting to image                                           0.1s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:3217a4a52760ed7b984ca6d6c427366f4ffdbd14b06d9d9d792ae059315bb9a1 0.0s
=> => exporting config sha256:51da7abdda96b493417b7d615703b857954a4ad222b624e51ae3930b11f5a60a 0.0s
=> => exporting attestation manifest sha256:fb2a467026e1667fe98445a9d26760c04f3d48f8a4a1aa2c8d6d7b2cc08b5308 0.0s
=> => exporting manifest list sha256:ff1a1afce84c67666f14e0cd0d80eed2011b9f32bc8c80381d5649be755340e9 0.0s
=> => naming to docker.io/library/marked-test:latest           0.0s
=> => unpacking to docker.io/library/marked-test:latest        0.0s
○ skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$

```

```

○ skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$ docker run --rm marked-test

> marked@17.0.5 test
> npm run build:reset && npm run build:docs && npm run test:specs && npm run test:unit && npm run test:umd && npm run test:cjs && npm run test:types && npm run test:lint

> marked@17.0.5 build:reset
> rimraf ./lib ./public

> marked@17.0.5 build:docs
> npm run build && node docs/build.js

> marked@17.0.5 build
> npm run build:esbuild && npm run build:types && npm run build:man

> marked@17.0.5 build:esbuild
> node esbuild.config.js

building version: 17.0.5

> marked@17.0.5 build:types
> tsc && dts-bundle-generator --export-referenced-types --project tsconfig.json -o lib/marked.d.ts src/marked.ts

Compiling input files...

```

```

✓marked unit (203.205852ms)
i tests 165
i suites 36
i pass 165
i fail 0
i cancelled 0
i skipped 0
i todo 0
i duration_ms 1446.46874

> marked@17.0.5 test:umd
> node test/umd-test.js

> marked@17.0.5 test:cjs
> node test/cjs-test.cjs

> marked@17.0.5 test:types
> tsc --project tsconfig-type-test.json && attw -P --entrypoints . --profile esm-only

marked v17.0.5

Build tools:
- @arethetypeswrong/cli@^0.18.2
- typescript@5.9.3
- esbuild@^0.27.4

(ignoring resolutions: 'node10', 'node16-cjs')

(ignored per resolution) ⚠A require call resolved to an ESM JavaScript file, which is an error in Node and some bundlers. CommonJS consumers will need to use a
dynamic import. https://github.com/arethetypeswrong/arethetypeswrong.github.io/blob/main/docs/problems/CJSResolvesToESM.md

```

|                   | "marked"                             |
|-------------------|--------------------------------------|
| node16 (from ESM) | ● (ESM)                              |
| bundler           | ●                                    |
| node10            | (ignored) ●                          |
| node16 (from CJS) | (ignored) ⚠ESM (dynamic import only) |

```

> marked@17.0.5 test:lint
> eslint

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$

```

Kod pliku konfiguracyjnego Docker Compose, ujmującego stworzone kontenery.

```

ITE > GCL3 > SK419875 > lab3 > docker-compose.yml
Run All Services
1 services:
  Run Service
2   test-runner:
3     build:
4       context: .
5       dockerfile: Dockerfile.test
6     container_name: marked-automated-test

```

```

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$ docker compose up --build
[+] Building 0.4s (7/7) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 579B 0.0s
=> [internal] load build definition from Dockerfile.test 0.0s
=> => transferring dockerfile: 146B 0.0s
=> [internal] load metadata for docker.io/library/marked-build:latest 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [1/1] FROM docker.io/library/marked-build:latest@sha256:c4010b4020e20ddc86c806657691634942097173221dc20 0.0s
=> => resolve docker.io/library/marked-build:latest@sha256:c4010b4020e20ddc86c806657691634942097173221dc20dfa9663 0.0s
=> exporting to image 0.1s
=> => exporting layers 0.0s
=> => exporting manifest sha256:38f75a68a7800f33e4f321bc2958ea718c3d06052ef9c701c7001beda855a5a2 0.0s
=> => exporting config sha256:73d74635846baf3e30514e29c2af91448e5ed9297db961758ac1f404010bab11 0.0s
=> => exporting attestation manifest sha256:2a428fad8a6d44ff4010ccae438e0c7f315142eefb35a48d278027e9099ba9ab 0.0s
=> => exporting manifest list sha256:e8311b476a65b49a23ba5f42a0a640c84a9e41c625b8a46ea9b9e7e7bdf8e80c 0.0s
=> => naming to docker.io/library/lab3-test-runner:latest 0.0s
=> => unpacking to docker.io/library/lab3-test-runner:latest 0.0s
=> resolving provenance for metadata file 0.0s
[+] up 3/3
✓ Image lab3-test-runner Built 0.5s
✓ Network lab3_default Created 0.1s
✓ Container marked-automated-test Created 0.1s
Attaching to marked-automated-test
marked-automated-test |
marked-automated-test | > marked@17.0.5 test
marked-automated-test | > npm run build:reset && npm run build:docs && npm run test:specs && npm run test:unit && npm ru

```

Stan Dockera po tych operacjach.

```

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker system df -v
Images space usage:

REPOSITORY          TAG          IMAGE ID       CREATED        SIZE      SHARED SIZE  UNIQUE SIZE  CONTAINERS
lab3-test-runner     latest      e8311b476a65  23 minutes ago  2.03GB    1.497GB      537.2MB      1
marked-build         latest      c4010b4020e2  23 minutes ago  2.03GB    1.497GB      537.2MB      0
marked-test         latest      ff1a1afce84c  23 minutes ago  2.03GB    1.497GB      537.2MB      0
node                20         e391c5561646  7 days ago     1.58GB    1.162GB      414.5MB      0
alpine/git          latest      d453f54c8332  7 weeks ago    141MB     0B           140.8MB      0

Containers space usage:

CONTAINER ID   IMAGE          COMMAND                  LOCAL VOLUMES   SIZE      CREATED        STATUS
NAMES
58a52ec79d5b   lab3-test-runner "docker-entrypoint.s..." 0               1.53MB     12 minutes ago Exited (0) 11 minut
es ago marked-automated-test

Local Volumes space usage:

VOLUME NAME    LINKS    SIZE
vol_enter      0        0B
vol_exit       0        0B

Build cache usage: 872.2MB

CACHE ID       CACHE TYPE    SIZE    CREATED        LAST USED        USAGE    SHARED
o1cqvgizso5i  regular      48.5MB  23 minutes ago  22 minutes ago  2        true
m2sa14c09e3j  regular      24MB    23 minutes ago  22 minutes ago  2        true
ibz1kdq599f3  regular      64.4MB  23 minutes ago  22 minutes ago  2        true
yavw8gc07yw5  regular      212MB   23 minutes ago  22 minutes ago  2        true
kbqmnkjdk4oa  regular      3.32kB  23 minutes ago  22 minutes ago  2        true
vk4vp73aihdd  regular      48.6MB  23 minutes ago  22 minutes ago  2        true
74bknx8xif3x  regular      1.25MB  23 minutes ago  22 minutes ago  2        true
pvm9n79e7dqu  regular      446B    23 minutes ago  22 minutes ago  2        true
u1ie3tokv3uv  regular      93B     23 minutes ago  22 minutes ago  2        true
6g0gohgipxk0  regular      25.6MB  23 minutes ago  22 minutes ago  2        false
xjmw9zfzg0gfa  regular      448MB   23 minutes ago  22 minutes ago  2        false
boduary666w4   source.local  4.1kB   23 minutes ago  12 minutes ago  3        false
wi25s1n98y61  source.local  0B      23 minutes ago  12 minutes ago  3        false
rlol9j9ajnc1  regular      564kB   23 minutes ago  12 minutes ago  3        false
skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ cd ../lab3
skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab3$ docker-compose down
[+] down 2/2
✓ Container marked-automated-test Removed 0.0s
✓ Network lab3_default Removed 0.1s

```

## DYSKUSJA

Czy program nadaje się do wdrażania i publikowania jako kontener, czy taki sposób Interakcji nadaje się tylko do builda? Opisz w jaki sposób miałyby zachodzić przygotowanie finalnego artefaktu.

- Zależy od programu, np. w przypadku biblioteki takiej jak moja - Marked.js, kontener służy głównie jako powtarzalne, odizolowane środowisko do bezpiecznego budowania i testowania. Sama biblioteka JS nie działa jako samodzielny proces produkcyjny (jak np. serwer www), więc nie ma po co wdrażać jej jako chodzącego kontenera, a raczej opublikować jej skompilowaną wersję.

Jeżeli program miałby być publikowany jako kontener - czy trzeba go oczyszczać z pozostałości po buildzie?

- Tak, bo końcowy obraz produkcyjny musi być jak najmniejszy (dla szybkości przesyłania) i bezpieczny. Należy stosować budowanie multi-stage build, aby w gotowym obrazie znalazł się tylko skompilowany kod, bez kompilatorów, kodu źródłowego, itp.

A może dedykowany deploy-and-publish byłby oddzielną ścieżką (inne Dockerfiles)?

- Tak, w architekturze produkcyjnej tworzy się oddzielny obraz docelowy (u mnie np. Alpine). Kopiuje się do niego wyłącznie czyste, gotowe artefakty pobrane z "brudnego" obrazu buildera.

Czy zbudowany program należałoby dystrybuować jako pakiet, np. JAR, DEB, RPM, EGG?

- Tak, dla bibliotek takich jak Marked.js, najbardziej naturalną formą ostatecznej dystrybucji jest pakiet w menedżerze NPM. Dla innych technologii byłyby to pakiety JAR (dla Javy) lub paczki DEB/RPM.

W jaki sposób zapewnić taki format? Dodatkowy krok (trzeci kontener)? Jakiś przykład?

- Wymaga to dodania kolejnego kroku na samym końcu Pipeline'u. Po udanym zbudowaniu i przejściu wszystkich testów, wywołuje się narzędzie, które loguje się do rejestru za pomocą specjalnego klucza i wypycha spakowany artefakt na serwer.

## LAB 4

Kopiowanie na wolumin wejściowy kontener bazowy z pomocą kontenera pomocniczego.

```

• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm -v vol_enter:/repo alpine/git clone https://github.com/markedsjs/markeds.git /repo/app
Cloning into '/repo/app'...
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker system df -v
Images space usage:

REPOSITORY          TAG          IMAGE ID      CREATED      SIZE      SHARED SIZE  UNIQUE SIZE  CONTAINERS
marked-test         latest       ff1a1afce84c  31 minutes ago  2.03GB    1.497GB      537.2MB      0
lab3-test-runner    latest       e8311b476a65  31 minutes ago  2.03GB    1.497GB      537.2MB      0
marked-build        latest       c4010b4020e2  31 minutes ago  2.03GB    1.497GB      537.2MB      0
node                20          e391c5561646  7 days ago    1.58GB    1.162GB      414.5MB      0
alpine/git          latest       d453f54c8332  7 weeks ago    141MB     0B           140.8MB      0

Containers space usage:

CONTAINER ID  IMAGE  COMMAND  LOCAL VOLUMES  SIZE  CREATED  STATUS  NAMES

Local Volumes space usage:

VOLUME NAME  LINKS  SIZE
vol_enter    0      12.72MB
vol_exit     0      0B

Build cache usage: 872.2MB

CACHE ID      CACHE TYPE  SIZE  CREATED      LAST USED      USAGE  SHARED
o1cqvgizso5i  regular    48.5MB  32 minutes ago  30 minutes ago  2      true
m2sa14c09e3j  regular    24MB    32 minutes ago  30 minutes ago  2      true
ibz1kdq599f3  regular    64.4MB  32 minutes ago  30 minutes ago  2      true
yaww8gc07yw5  regular    212MB   32 minutes ago  30 minutes ago  2      true
kbqmnkjdk4oa  regular    3.32kB  32 minutes ago  30 minutes ago  2      true
vk4vp73aihdd  regular    48.6MB  32 minutes ago  30 minutes ago  2      true
74bkx8xif3x   regular    1.25MB  32 minutes ago  30 minutes ago  2      true
pvm9n79e7dqu  regular    446B    32 minutes ago  30 minutes ago  2      true
u1ie3tokv3uv  regular    93B     32 minutes ago  30 minutes ago  2      true
6g0gohgipxk0  regular    25.6MB  31 minutes ago  30 minutes ago  2      false
xjmw9zfg0gfa  regular    448MB   31 minutes ago  30 minutes ago  2      false
boduery666w4  source.local 4.1kB   32 minutes ago  20 minutes ago  3      false
wi25sln98y61  source.local 0B       32 minutes ago  20 minutes ago  3      false
r1ol9j9ajnc1  regular    564kB   31 minutes ago  20 minutes ago  3      false
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm -v vol_enter:/repo -v vol_exit:/out -w /repo/app node:20 bash -c "npm install && npm run build && cp -r . /out/"
npm warn EBADENGINE Unsupported engine {
npm warn EBADENGINE   package: '@semantic-release/github@12.0.6',
npm warn EBADENGINE   required: { node: '^22.14.0 || >= 24.10.0' },
npm warn EBADENGINE   current: { node: 'v20.20.1', npm: '10.8.2' }
npm warn EBADENGINE }
npm warn EBADENGINE Unsupported engine {
npm warn EBADENGINE   package: '@semantic-release/npm@13.1.5',
npm warn EBADENGINE   required: { node: '^22.14.0 || >= 24.10.0' },
npm warn EBADENGINE   current: { node: 'v20.20.1', npm: '10.8.2' }

```

Build w kontenerze bazowym i zapis na wolumin wyjściowy.

```

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm -v vol_enter:/repo -v vol_exit:/out -w /repo/app node:20 bash -c "npm install && npm run build && cp -r . /out/"
npm warn EBADENGINE Unsupported engine {
  npm warn EBADENGINE   package: '@semantic-release/github@12.0.6',
  npm warn EBADENGINE   required: { node: '^22.14.0 || >= 24.10.0' }, ...
}
Processing src/marked.ts
Writing src/marked.ts -> lib/marked.d.ts
Checking generated files...
Done in 10.48s

> marked@17.0.5 build:man
> marked-man man/marked.1.md > man/marked.1

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker system df -v
Images space usage:

REPOSITORY          TAG          IMAGE ID      CREATED      SIZE      SHARED SIZE  UNIQUE SIZE  CONTAINERS
lab3-test-runner    latest      e8311b476a65 33 minutes ago 2.03GB     1.497GB      537.2MB      0
marked-build        latest      c4010b4020e2 33 minutes ago 2.03GB     1.497GB      537.2MB      0
marked-test         latest      ff1a1afce84c 33 minutes ago 2.03GB     1.497GB      537.2MB      0
node                20         e391c5561646 7 days ago    1.58GB     1.162GB      414.5MB      0
alpine/git          latest      d453f54c8332 7 weeks ago   141MB      0B           140.8MB      0

Containers space usage:

CONTAINER ID        IMAGE          COMMAND        LOCAL VOLUMES   SIZE      CREATED        STATUS        NAMES
Local Volumes space usage:

VOLUME NAME        LINKS         SIZE
vol_exit            0            231.3MB
vol_enter           0            220.2MB

Build cache usage: 872.2MB

CACHE ID           CACHE TYPE     SIZE      CREATED      LAST USED     USAGE     SHARED
o1cqvgizso5i      regular        48.5MB    33 minutes ago 32 minutes ago 2         true
m2sa14c09e3j      regular        24MB      33 minutes ago 32 minutes ago 2         true
ibzlkdq599f3      regular        64.4MB    33 minutes ago 32 minutes ago 2         true
yavw8gc07yw5      regular        212MB     33 minutes ago 32 minutes ago 2         true
kbqmnkjkd4oa      regular        3.32kB    33 minutes ago 32 minutes ago 2         true
vk4vp73aihdd      regular        48.6MB    33 minutes ago 32 minutes ago 2         true
74bkx8xif3x       regular        1.25MB    33 minutes ago 32 minutes ago 2         true
pvm9n79e7dqu      regular        446B      33 minutes ago 32 minutes ago 2         true
u1ie3tokv3uv      regular        93B       33 minutes ago 32 minutes ago 2         true
6g0gohgipxk0      regular        25.6MB    33 minutes ago 32 minutes ago 2         false
xjmw9zfg0gfa      regular        448MB     33 minutes ago 32 minutes ago 2         false
boduery666w4      source.local   4.1kB     33 minutes ago 22 minutes ago 3         false
wi25sln98y61      source.local   0B        33 minutes ago 22 minutes ago 3         false
r1ol9j9ajnc1      regular        564kB     33 minutes ago 22 minutes ago 3         false

```

Wykorzystuję Alpine.

```

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm -v vol_exit:/out alpine ls -la /out
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
caa817ad3aea: Download complete
9e595aac14e0: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
total 388
drwxr-xr-x 13 root    root          4096 Mar 24 16:07 .
drwxr-xr-x  1 root    root           17 Mar 24 16:11 ..
drwxr-xr-x  2 root    root           31 Mar 24 16:07 .devcontainer
-rw-r--r--  1 root    root          272 Mar 24 16:07 .editorconfig
drwxr-xr-x  7 root    root           147 Mar 24 16:07 .git
-rw-r--r--  1 root    root           76 Mar 24 16:07 .gitattributes
drwxr-xr-x  5 root    root          153 Mar 24 16:07 .github
-rw-r--r--  1 root    root          116 Mar 24 16:07 .gitignore
-rw-r--r--  1 root    root          202 Mar 24 16:07 .releaserc.json
-rw-r--r--  1 root    root           48 Mar 24 16:07 CHANGELOG.md
-rw-r--r--  1 root    root          2942 Mar 24 16:07 LICENSE.md
-rw-r--r--  1 root    root          3116 Mar 24 16:07 README.md
-rw-r--r--  1 root    root           554 Mar 24 16:07 SECURITY.md
drwxr-xr-x  2 root    root           23 Mar 24 16:07 api
drwxr-xr-x  2 root    root           38 Mar 24 16:07 bin
drwxr-xr-x  6 root    root          4096 Mar 24 16:07 docs
-rw-r--r--  1 root    root          1117 Mar 24 16:07 esbuild.config.js
-rw-r--r--  1 root    root          173 Mar 24 16:07 eslint.config.js
drwxr-xr-x  2 root    root           117 Mar 24 16:07 lib
drwxr-xr-x  2 root    root           41 Mar 24 16:07 man
drwxr-xr-x 342 root    root          12288 Mar 24 16:07 node_modules
-rw-r--r--  1 root    root        306541 Mar 24 16:07 package-lock.json
-rw-r--r--  1 root    root          3396 Mar 24 16:07 package.json
drwxr-xr-x  2 root    root          4096 Mar 24 16:07 src
drwxr-xr-x  5 root    root          4096 Mar 24 16:07 test
-rw-r--r--  1 root    root          198 Mar 24 16:07 tsconfig-type-test.json
-rw-r--r--  1 root    root          360 Mar 24 16:07 tsconfig.json
-rw-r--r--  1 root    root           24 Mar 24 16:07 vercel.json

```

```

● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker system df -v
Images space usage:

REPOSITORY          TAG          IMAGE ID       CREATED        SIZE      SHARED SIZE  UNIQUE SIZE   CONTAINERS
lab3-test-runner    latest      e8311b476a65  37 minutes ago  2.03GB    1.497GB      537.2MB      0
marked-build        latest      c4010b4020e2  37 minutes ago  2.03GB    1.497GB      537.2MB      0
marked-test         latest      ff1a1afce84c  37 minutes ago  2.03GB    1.497GB      537.2MB      0
node                 20         e391c5561646  7 days ago     1.58GB    1.162GB      414.5MB      0
alpine/git          latest      d453f54c8332  7 weeks ago    141MB     8.741MB      132.1MB      0
alpine              latest      25109184c71b  7 weeks ago    12.7MB    8.741MB      3.95MB       0

Containers space usage:

CONTAINER ID        IMAGE               COMMAND          LOCAL VOLUMES   SIZE      CREATED        STATUS        NAMES

Local Volumes space usage:

VOLUME NAME        LINKS     SIZE
vol_enter           0         220.2MB
vol_exit            0         231.3MB

Build cache usage: 872.2MB

CACHE ID            CACHE TYPE        SIZE      CREATED        LAST USED        USAGE     SHARED
o1cqvgizso5i        regular           48.5MB    37 minutes ago  36 minutes ago   2         true
m2sa14c09e3j        regular           24MB      37 minutes ago  36 minutes ago   2         true
ibz1kdq599f3        regular           64.4MB    37 minutes ago  36 minutes ago   2         true
yavw8gc07yw5        regular           212MB     37 minutes ago  36 minutes ago   2         true
kbqmnkjdk4oa        regular           3.32kB    37 minutes ago  36 minutes ago   2         true
vk4vp73aihdd        regular           48.6MB    37 minutes ago  36 minutes ago   2         true
74bknx8xf3x         regular           1.25MB    37 minutes ago  36 minutes ago   2         true
pvm9n79e7dqu        regular           446B      37 minutes ago  36 minutes ago   2         true
utie3tokv3uv        regular           93B       37 minutes ago  36 minutes ago   2         true
6g0gohgipxk0        regular           25.6MB    37 minutes ago  36 minutes ago   2         false
xjmw9zfg0gfa        regular           448MB     37 minutes ago  36 minutes ago   2         false
boduary666w4        source.local      4.1kB     37 minutes ago  26 minutes ago   3         false
wi25sln98y61        source.local      0B        37 minutes ago  26 minutes ago   3         false
rlol9j9ajnc1        regular           564kB     37 minutes ago  26 minutes ago   3         false

```

Sposób alternatywny: uruchomienie node:20 i instalacja gita i pobranie repo.

```
● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm -v vol_enter_git:/app -w /app node:20 bash
-c "apt-get update && apt-get install -y git && git clone https://github.com/arkedjs/arked.git . && ls -la"
Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [294 kB]
Fetched 9347 kB in 2s (5953 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
git is already the newest version (1:2.39.5-0+deb12u3).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Cloning into '.'...
total 372
drwxr-xr-x. 11 root root 4096 Mar 24 16:18 .
drwxr-xr-x. 1 root root 39 Mar 24 16:17 ..
drwxr-xr-x. 2 root root 31 Mar 24 16:18 .devcontainer
-rw-r--r--. 1 root root 272 Mar 24 16:18 .editorconfig
drwxr-xr-x. 8 root root 163 Mar 24 16:18 .git
-rw-r--r--. 1 root root 76 Mar 24 16:18 .gitattributes
drwxr-xr-x. 5 root root 153 Mar 24 16:18 .github
-rw-r--r--. 1 root root 116 Mar 24 16:18 .gitignore
-rw-r--r--. 1 root root 202 Mar 24 16:18 .releaserc.json
-rw-r--r--. 1 root root 48 Mar 24 16:18 CHANGELOG.md
-rw-r--r--. 1 root root 2942 Mar 24 16:18 LICENSE.md
-rw-r--r--. 1 root root 3116 Mar 24 16:18 README.md
-rw-r--r--. 1 root root 554 Mar 24 16:18 SECURITY.md
drwxr-xr-x. 2 root root 23 Mar 24 16:18 api
drwxr-xr-x. 2 root root 38 Mar 24 16:18 bin
drwxr-xr-x. 6 root root 4096 Mar 24 16:18 docs
-rw-r--r--. 1 root root 1117 Mar 24 16:18 esbuild.config.js
-rw-r--r--. 1 root root 173 Mar 24 16:18 eslint.config.js
drwxr-xr-x. 2 root root 25 Mar 24 16:18 man
-rw-r--r--. 1 root root 306541 Mar 24 16:18 package-lock.json
-rw-r--r--. 1 root root 3396 Mar 24 16:18 package.json
drwxr-xr-x. 2 root root 4096 Mar 24 16:18 src
drwxr-xr-x. 5 root root 4096 Mar 24 16:18 test
-rw-r--r--. 1 root root 198 Mar 24 16:18 tsconfig-type-test.json
-rw-r--r--. 1 root root 360 Mar 24 16:18 tsconfig.json
-rw-r--r--. 1 root root 24 Mar 24 16:18 vercel.json
● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$
```

Pobranie iperf3 i odpalenie serwera, pobranie adresu serwera, uruchomienie klienta.

```

• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run -d --rm --name serwer_iperf alpine sh -c "apk up
date && apk add iperf3 && iperf3 -s"
351fdee45de3dc7ceb373d418556e45049354e86cc02fd366aacbf90bcb87e74
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPA
ddress}}{{end}}' serwer_iperf
172.17.0.2
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm alpine sh -c "apk update && apk add iperf3
&& iperf3 -c 172.17.0.2"
v3.23.3-358-gbe4aca58ed8 [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.3-357-gba1e13dee22 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27575 distinct packages available
(1/1) Installing iperf3 (3.19.1-r1)
Executing busybox-1.37.0-r30.trigger
OK: 8412 KiB in 17 packages
Connecting to host 172.17.0.2, port 5201
[ 5] local 172.17.0.3 port 53252 connected to 172.17.0.2 port 5201
[ ID] Interval          Transfer    Bitrate      Retr  Cwnd
[ 5] 0.00-1.00 sec      3.11 GBytes 26.7 Gbits/sec  0    3.99 MBytes
[ 5] 1.00-2.00 sec      3.11 GBytes 26.7 Gbits/sec  0    3.99 MBytes
[ 5] 2.00-3.00 sec      3.22 GBytes 27.6 Gbits/sec  0    3.99 MBytes
[ 5] 3.00-4.00 sec      2.78 GBytes 23.9 Gbits/sec  0    3.99 MBytes
[ 5] 4.00-5.00 sec      2.96 GBytes 25.4 Gbits/sec  0    3.99 MBytes
[ 5] 5.00-6.00 sec      3.19 GBytes 27.4 Gbits/sec  0    3.99 MBytes
[ 5] 6.00-7.00 sec      3.25 GBytes 27.9 Gbits/sec  0    3.99 MBytes
[ 5] 7.00-8.00 sec      3.28 GBytes 28.1 Gbits/sec  0    3.99 MBytes
[ 5] 8.00-9.00 sec      3.22 GBytes 27.6 Gbits/sec  0    3.99 MBytes
[ 5] 9.00-10.00 sec     3.12 GBytes 26.8 Gbits/sec  0    3.99 MBytes
-----
[ ID] Interval          Transfer    Bitrate      Retr
[ 5] 0.00-10.00 sec    31.2 GBytes 26.8 Gbits/sec  0
[ 5] 0.00-10.00 sec    31.2 GBytes 26.8 Gbits/sec  0
sender
receiver

iperf Done.

```

Stworzenie wirtualnej sieci, uruchomienie serwera w nowej sieci i uruchomienie klienta za pomocą nazwy (a nie IP).

```

• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker network create moja_siec
e5198bf8a97224dbb2e16ee2b064a43527406e6f6e67fcf8a15611751a6c38aa
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker stop serwer_iperf
serwer_iperf
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run -d --rm --name serwer_iperf_dns --network moja_s
iec alpine sh -c "apk update && apk add iperf3 && iperf3 -s"
f75da46a85f9f90244bd7c68f3420f6b2d08d8859e4074d0de13f5739b3faa08
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run --rm --network moja_siec alpine sh -c "apk updat
e && apk add iperf3 && iperf3 -c serwer_iperf_dns"
v3.23.3-358-gbe4aca58ed8 [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.3-357-gba1e13dee22 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27575 distinct packages available
(1/1) Installing iperf3 (3.19.1-r1)
Executing busybox-1.37.0-r30.trigger
OK: 8412 KiB in 17 packages
Connecting to host serwer_iperf_dns, port 5201
[ 5] local 172.18.0.3 port 54030 connected to 172.18.0.2 port 5201
[ ID] Interval          Transfer    Bitrate      Retr  Cwnd
[ 5] 0.00-1.00 sec      2.97 GBytes 25.5 Gbits/sec  0    4.13 MBytes
[ 5] 1.00-2.00 sec      3.00 GBytes 25.8 Gbits/sec  0    4.13 MBytes
[ 5] 2.00-3.00 sec      3.05 GBytes 26.2 Gbits/sec  0    4.13 MBytes
[ 5] 3.00-4.00 sec      2.97 GBytes 25.5 Gbits/sec  0    4.13 MBytes
[ 5] 4.00-5.00 sec      3.01 GBytes 25.8 Gbits/sec  0    4.13 MBytes
[ 5] 5.00-6.00 sec      2.61 GBytes 22.5 Gbits/sec  2    4.13 MBytes
[ 5] 6.00-7.00 sec      2.70 GBytes 23.2 Gbits/sec  0    4.13 MBytes
[ 5] 7.00-8.00 sec      2.58 GBytes 22.1 Gbits/sec  1    4.13 MBytes
[ 5] 8.00-9.00 sec      2.82 GBytes 24.2 Gbits/sec  0    4.13 MBytes
[ 5] 9.00-10.00 sec     2.76 GBytes 23.7 Gbits/sec  0    4.13 MBytes
-----
[ ID] Interval          Transfer    Bitrate      Retr
[ 5] 0.00-10.00 sec    28.5 GBytes 24.4 Gbits/sec  3
[ 5] 0.00-10.00 sec    28.5 GBytes 24.4 Gbits/sec  0
sender
receiver

iperf Done.
• skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$

```

Łączenie się z serwerem spoza kontenera.

```

● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker stop serwer_iperf_dns
serwer_iperf_dns
● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker run -d --rm --name serwer_iperf_host -p 5201:5201 al
pine sh -c "apk update && apk add iperf3 && iperf3 -s"
07197e3e9f74168667585d0101815054a9004d55f1cfd56da0eb20eb7e6014c
● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ sudo dnf install -y iperf3
[sudo] password for skrzysztofik:
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
iperf3 x86_64 3.19.1-1.fc43 fedora 301.5 KiB
Installing dependencies:
lksctp-tools x86_64 1.0.21-1.fc43 fedora 251.0 KiB

Transaction Summary:
Installing: 2 packages

Total size of inbound packages is 220 KiB. Need to download 220 KiB.
After this operation, 552 KiB extra will be used (install 552 KiB, remove 0 B).
[1/2] iperf3-0:3.19.1-1.fc43.x86_64 100% | 194.5 KiB/s | 123.7 KiB | 00m01s
[2/2] lksctp-tools-0:1.0.21-1.fc43.x86_64 100% | 134.0 KiB/s | 96.8 KiB | 00m01s
-----
[2/2] Total 100% | 223.1 KiB/s | 220.5 KiB | 00m01s
Running transaction
[1/4] Verify package files 100% | 333.0 B/s | 2.0 B | 00m00s
[2/4] Prepare transaction 100% | 18.0 B/s | 2.0 B | 00m00s
[3/4] Installing lksctp-tools-0:1.0.21-1.fc43.x86_64 100% | 6.6 MiB/s | 255.3 KiB | 00m00s
[4/4] Installing iperf3-0:3.19.1-1.fc43.x86_64 100% | 995.5 KiB/s | 303.6 KiB | 00m00s
Complete!
● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ iperf3 -c 127.0.0.1
Connecting to host 127.0.0.1, port 5201
[ 5] local 127.0.0.1 port 53788 connected to 127.0.0.1 port 5201
[ ID] Interval Transfer Bitrate Retr Cwnd
[ 5] 0.00-1.01 sec 2.05 GBytes 17.5 Gbits/sec 0 4.12 MBytes
[ 5] 1.01-2.00 sec 2.26 GBytes 19.5 Gbits/sec 0 4.12 MBytes
[ 5] 2.00-3.00 sec 2.11 GBytes 18.1 Gbits/sec 0 4.12 MBytes
[ 5] 3.00-4.00 sec 1.85 GBytes 15.8 Gbits/sec 0 4.12 MBytes
[ 5] 4.00-5.00 sec 2.10 GBytes 18.1 Gbits/sec 0 4.12 MBytes
[ 5] 5.00-6.00 sec 2.16 GBytes 18.5 Gbits/sec 0 4.12 MBytes
[ 5] 6.00-7.00 sec 2.23 GBytes 19.1 Gbits/sec 0 4.12 MBytes
[ 5] 7.00-8.00 sec 2.25 GBytes 19.4 Gbits/sec 0 4.12 MBytes
[ 5] 8.00-9.00 sec 2.26 GBytes 19.4 Gbits/sec 0 4.12 MBytes
[ 5] 9.00-10.00 sec 2.33 GBytes 20.0 Gbits/sec 0 4.12 MBytes
-----
[ ID] Interval Transfer Bitrate Retr sender receiver
[ 5] 0.00-10.00 sec 21.6 GBytes 18.6 Gbits/sec 0
[ 5] 0.00-10.00 sec 21.5 GBytes 18.5 Gbits/sec

iperf Done.
○ skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$

```

Serwer SSH.

```
^Cskrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab4$ ssh -p 2222 root@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:j8CdHw7ahBPdF+5UL3ALo2uAfx+A/A60Ggw/bd8A8U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
root@127.0.0.1's password:
Permission denied, please try again.
root@127.0.0.1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.13-200.fc43.x86_64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@714c1aec0cfe:~#
```

Jenkins i DIND.

```

● skrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab4$ docker network create jenkins
b3e2f92383b018bc08e96dc0331540e57947ed1c6bfa81706ac130bdb6063d53
● skrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab4$ docker run --name jenkins-docker --rm --detach \
--privileged --network jenkins --network-alias docker \
--env DOCKER_TLS_CERTDIR=/certs \
--volume jenkins-docker-certs:/certs/client \
--volume jenkins-data:/var/jenkins_home \
docker:dind
Unable to find image 'docker:dind' locally
dind: Pulling from library/docker
31bd52174687: Pull complete
a524d6d9cccd: Pull complete
8b4da28e0bc4: Pull complete
503f0e2071dd: Pull complete
bcdcbaa55207: Pull complete
d3f1c7a58cda: Pull complete
4f4fb700ef54: Pull complete
7ba58ca1e42e: Pull complete
12f8c3243c30: Pull complete
89e9c4ffbcbf: Pull complete
34962ed90623: Pull complete
04a9731c772a: Pull complete
a4a294ee3d24: Pull complete
9df03e33ed49: Pull complete
92fbd819f55b: Pull complete
9de809ee0c1e: Download complete
Digest: sha256:1ba18449911d01f477a3fc104123c85d677addc60701b14b3fcb5381f9c4a537
Status: Downloaded newer image for docker:dind
8801d4913a32980ddb6c534e6651c098753fb281c9f75d5a2919e456d30922f0
● skrzysztofik@localhost:~/MD02026s_ITE/ITE/GCL3/SK419875/lab4$ docker run --name jenkins-server --rm --detach \
--network jenkins --env DOCKER_HOST=tcp://docker:2376 \
--env DOCKER_CERT_PATH=/certs/client --env DOCKER_TLS_VERIFY=1 \
--volume jenkins-data:/var/jenkins_home \
--volume jenkins-docker-certs:/certs/client:ro \
--publish 8080:8080 --publish 50000:50000 \
jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
e682a95ad036: Pull complete
53c88f1dfeb7: Pull complete
4d2125ccf71c: Pull complete
69503609dc01: Pull complete
6da7907f70f4: Pull complete
6327a7366a6f: Pull complete
01938f7c040b: Pull complete
b7b154fd69a1: Pull complete
876217fdee20: Pull complete
fbae626c1b7a: Pull complete
b19825c62ed8: Pull complete
ec7478903548: Pull complete
a7f97ef35fd8: Download complete
Digest: sha256:c4098086090ca98491d4bf66182f5e3b015a8232f2acf2df209a212a5801aa8e
Status: Downloaded newer image for jenkins/jenkins:lts
a6605c8569d8dffae3b7b4636acb6af42d53e9dffd893a3824ea4bb70242bb

```

```

● skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker system df -v
Images space usage:

REPOSITORY          TAG          IMAGE ID       CREATED        SIZE      SHARED SIZE  UNIQUE SIZE  CONTAINERS
marked-build        latest       c4010b4020e2   About an hour ago  2.03GB    1.497GB      537.2MB      0
marked-test         latest       ff1a1afce84c   About an hour ago  2.03GB    1.497GB      537.2MB      0
lab3-test-runner    latest       e8311b476a65   About an hour ago  2.03GB    1.497GB      537.2MB      0
jenkins/jenkins     lts         c4098086090c   6 days ago      800MB     0B           799.6MB      1
node                20          e391c5561646   7 days ago      1.58GB    1.162GB      414.5MB      0
docker              dind        1ba18449911d   2 weeks ago     542MB     8.741MB      533.7MB      1
ubuntu              22.04       ce4a593b4e32   4 weeks ago     116MB     0B           116.4MB      0
alpine/git          latest      d453f54c8332   7 weeks ago     141MB     8.741MB      132.1MB      0
alpine              latest      25109184c71b   7 weeks ago     12.7MB    8.741MB      3.95MB       0

Containers space usage:

CONTAINER ID   IMAGE          COMMAND                  LOCAL VOLUMES   SIZE      CREATED        STATUS
a6605c8569d8   jenkins/jenkins:lts   "/usr/bin/tini -- /u..."  2              4.93MB    13 seconds ago Up 10 seconds
jenkins-server
8801d4913a32   docker:dind        "dockerd-entrypoint..."  3              65.5kB    40 seconds ago Up 37 seconds
jenkins-docker

Local Volumes space usage:

VOLUME NAME                               LINKS   SIZE
vol_exit                                  0       231.3MB
jenkins-docker-certs                     2       8.586kB
jenkins-data                             2       108.3MB
7467fba5e39977bd08339a86469200dc3bef5247932eaa150b2ee1e67ca5b5a8  1       278.6kB
vol_enter_git                             0       12.66MB
vol_enter                                 0       220.2MB

Build cache usage: 872.2MB

CACHE ID      CACHE TYPE   SIZE      CREATED        LAST USED        USAGE    SHARED
o1cqvgizso5i  regular     48.5MB    About an hour ago  About an hour ago  2       true
m2sa14c09e3j  regular     24MB      About an hour ago  About an hour ago  2       true
ibz1kdq599f3  regular     64.4MB    About an hour ago  About an hour ago  2       true
yavw8gc07yw5  regular     212MB     About an hour ago  About an hour ago  2       true
kbqmnkjdk4oa  regular     3.32kB    About an hour ago  About an hour ago  2       true
vk4vp73aihdd  regular     48.6MB    About an hour ago  About an hour ago  2       true
74bkx8xi3x    regular     1.25MB    About an hour ago  About an hour ago  2       true
pvm9n79e7dqu  regular     446B      About an hour ago  About an hour ago  2       true
u1ie3tokv3uv  regular     93B       About an hour ago  About an hour ago  2       true
6g0gohgipxk0  regular     25.6MB    About an hour ago  About an hour ago  2       false
xjmw9zfg0gfa  regular     448MB     About an hour ago  About an hour ago  2       false
boduery666w4  source.local 4.1kB     About an hour ago  About an hour ago  3       false
wi25sln98y61  source.local 0B        About an hour ago  About an hour ago  3       false
r1o19j9ajnc1  regular     564kB     About an hour ago  About an hour ago  3       false

```

Hasło do panelu Jenkins.

```

skrzysztofik@localhost:~/MD02026s ITE/ITE/GCL3/SK419875/lab4$ docker logs jenkins-server
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-03-24 16:46:34.788+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-03-24 16:46:35.890+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath ...
22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 21.0.9+10-LTS
2026-03-24 16:46:36.840+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /,
did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-03-24 16:46:36.973+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-03-24 16:46:37.723+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkin
s_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-03-24 16:46:37.935+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$Cor
eContextHandler@3681037{Jenkins v2.541.3,/,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContex
tHandler$CoreToNestedHandler@2459319c{STARTED}}
2026-03-24 16:46:37.967+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@2aa6
311a{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-03-24 16:46:37.968+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@78525ef9{STA
RTING}[12.1.5,sto=0] @1969ms
2026-03-24 16:46:37.974+0000 [id=25] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort
=disabled
2026-03-24 16:46:38.141+0000 [id=24] INFO jenkins.model.Jenkins#<init>: Starting version 2.541.3
2026-03-24 16:46:38.286+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-03-24 16:46:38.317+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-03-24 16:46:40.140+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-03-24 16:46:40.150+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-03-24 16:46:40.153+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-03-24 16:46:40.568+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-03-24 16:46:40.569+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-03-24 16:46:40.570+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-03-24 16:46:40.573+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs update
d
2026-03-24 16:46:40.631+0000 [id=46] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates serv
er
2026-03-24 16:46:41.821+0000 [id=31] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 5521891788794f108d24329ba1b6388d
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
2026-03-24 16:46:49.407+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-03-24 16:46:49.450+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-03-24 16:46:50.492+0000 [id=46] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for
hudson.tasks.Maven.MavenInstaller
2026-03-24 16:46:50.493+0000 [id=46] INFO hudson.util.Retrier#start: Performed the action check updates server succ

```

## Getting Started

# Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

**Administrator password**

Continue

Jenkins w przeglądarce na adresie serwera.

**Jenkins**

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status 0/2

Add description

## Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

### Start building your software project

Create a job +

### Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.541.3

172.23.187.229:8080/view/all/newJob

## DYSKUSJA

Możliwość użycia komendy RUN --mount

- Instrukcja RUN --mount=type=bind tymczasowo montuje kod źródłowy podczas docker build bez trwałego kopiowania go do warstw obrazu instrukcja COPY. Oszczędza to pamięć, przyspiesza budowanie i eliminuje potrzebę kontenerów pomocniczych.

Wady i zalety SSH wewnątrz kontenera.

- Wady: Łamie to główną zasadę Dockera że jeden kontener to jeden proces. Wymusza instalację i utrzymywanie dodatkowych zasobów (SSHD) i drastycznie

zwiększa ryzyko ataków. Do administracji i podglądania kontenera służy polecenie `docker exec`, a nie SSH.

- Zalety: Pozwala traktować kontener w 100% jak pełnoprawną maszynę wirtualną, co jest przydatne tylko przy przenoszeniu bardzo starego, nieskonteneryzowanego poprawnie oprogramowania.